

EDUCATION

University of New South Wales

Master of Commerce in Business Analytics

Sydney, AU

Feb. 2026 - Dec. 2026 (Expected)

Hong Kong Baptist University

Bachelor of Business Administration (Honours) in Accounting

Hong Kong SAR

Sep. 2021 - Jun. 2025

- Studied at Beijing Normal-Hong Kong Baptist University (BNBU), Zhuhai Campus
- Second Class Honours

RESEARCH INTERESTS

- LLM evaluation
- Retrieval-augmented generation
- Knowledge graphs
- ESG and sustainability disclosure
- AI in accounting and finance

PUBLICATIONS & RESEARCH PAPERS

Ding, Y., Yang, Z., Lu, Y., Lan, L., Tang, Y., Yang, Z., Tang, X., **Hu, J.**, Ju, M., Zhang, W., Yang, W., Xu, X., & Zhang, W. (2026). EulerESG: A unified platform for standards-aware ESG disclosure analysis. In *Proceedings of the IEEE International Conference on Data Mining (ICDM), Demo Track*, Shenyang, China. (Accepted).

Hu, J., & Wang, Y. (2025). *The Impact of Corporate Environmental, Social and Governance (ESG) Disclosure on Corporate Investment Efficiency* [Unpublished undergraduate final year project]. Beijing Normal-Hong Kong Baptist University. [[PDF](#)] [[Code](#)]

RESEARCH EXPERIENCE

Research Assistant, Remote

Jun. 2026 - Present

Vecton AI Pty Ltd

Sydney, AU

- Working on retrieval-augmented generation and LLM agent development, progressing from baseline to advanced RAG (query rewriting, hybrid BM25 and vector retrieval, reranking, RAG evaluation) and on to tool calling, agent memory and multi-agent workflows, using FastAPI, LangChain and vector databases.
- Strengthening the model-level foundations behind this work, from tokenization, Transformer architecture and training systems to post-training, alongside hands-on study of attention and sampling in GPT-2.

Student Researcher, Remote

May. 2026 - Present

RAIDS Lab, University of Sydney

Sydney, AU

Supervisor: Dr. Zhengyi Yang [[Details](#)]

- Conduct research on standards-aware ESG disclosure analysis and the reliability of large-language-model judgements as measurement.
- Extended an ESG extraction and scoring workflow, analyzed how item-level discrepancies propagate into report-level scores and rankings, and contributed to the EulerESG manuscript and system review.
- Collaborating with Zherui Wang (UNSW CSE) on reproducible diagnostics and independently annotated benchmark design for ESG disclosure assessment; detailed below as a separate research project.
- Developing an ontology-guided research proposal on evidence grounding and cross-dimension bias in LLM-generated ESG assessments.

SELECTED PROJECTS

Vecton AI – Business Analysis & Financial Planning

China International College Students' Innovation Competition 2026

Sep. 2026 - Present

- Built consolidated three-year financial statements for the Australian and Chinese entities.
- Added break-even, payment-term and dilution scenarios to size funding needs, presented in bilingual pitch decks.

Reproducible Diagnostics and Benchmark Design for ESG Disclosure Assessment

UNSW, Sydney

Aug. 2026 - Present

Supervisor: Dr. Zhengyi Yang

- Compared human-reviewed labels with raw model outputs across 25,224 ESG disclosure records covering 37 ASX-listed entities and 53 report-years under AASB S1/S2, GRI and SASB; human review added evidence conditions in 5,310 cases and removed them in 1,211 cases.
- Found that evidence presence alone reproduced 99.92% of the corpus's disclosed/not-disclosed labels, motivating evaluation against individual disclosure requirements rather than aggregate labels alone.
- Built a reproducible diagnostic pipeline documenting 13 data-quality issues and assessed label balance across 29 AASB S2 greenhouse-gas requirements to select a benchmark subset.
- Designed a blind double-annotation protocol for 300 items, including annotator calibration, agreement assessment and adjudication.

Carbon Chain – Cross-Border Sustainability Compliance Verification

China International College Students' Innovation Competition 2027

Jun. 2026 - Present

Supervisor: Dr. Zhengyi Yang

- Lead a backend, frontend and evaluation team extending the EulerESG platform to Chinese disclosure rules.
- Designed and implemented a four-layer graph (ESG entities, claim-evidence, provenance, temporal) that scores how far a disclosure claim is supported by traceable evidence, counting evidence by independent source rather than by document.
- Collected 48 reports from 10 A-/H-share issuers (2021-2025) through CNINFO and piloted the workflow on CATL's carbon-neutrality disclosures, identifying restatements and differences in accounting standards as barriers to comparison.

ESG-Augmented Corporate Credit Risk Prediction and Explainability Analysis

2026, UNSW, Sydney [\[Code\]](#)

- Built an XGBoost pipeline predicting KMV distance-to-default on a 42,000+ firm-year A-share panel (2015-2025) with 18 financial, market and ESG features, reaching $R^2 = 0.68$, plus a distress classifier predicting next-year ST/*ST status at ROC-AUC = 0.92; avoided target leakage by rejecting Z-score for a market-based risk measure.
- Quantified the incremental value of ESG through a firm-grouped cross-validation ablation, isolating a small but statistically significant gain (-1.25% RMSE, $p = 0.003$) that held across three distance-to-default definitions and an independent distress target, with multi-layer SHAP explainability.

Multi-Factor Quantitative Stock Selection & Backtesting

2026, UNSW, Sydney [\[Code\]](#)

- Built a point-in-time factor-research pipeline on 498,893 stock-months across 5,546 A-share firms including delisted names (2015-2025), engineering 8 style and risk factors with winsorization, industry and size neutralization, and a statutory-disclosure lag to remove look-ahead bias.
- Evaluated factors via Spearman rank IC, ICIR and decay curves, surfacing a dominant negative turnover effect, small-cap and low-volatility premia, and statistically absent momentum; a walk-forward LightGBM composite reached ICIR 0.72 and a gross long-short quintile Sharpe of 0.93 versus 0.31 for the CSI 300, before transaction costs and A-share short-sale constraints.

Chinese White Dolphin Habitat Loss and Conservation in Western Hong Kong

2023, BNU [\[Code\]](#)

- Built a reproducible PDF-to-panel pipeline (PyMuPDF, pandas) parsing ten years of AFCD marine-mammal monitoring reports into a validated area $\times$ monitoring-period panel of 2,209 sightings and 7,023 survey-effort records, aligned with EPD station-level water-quality data.
- Applied a Negative Binomial GLM with area fixed effects and a survey-effort offset, kernel utilization distributions, a Poisson tensor-smooth spatial GAM, and Pettitt / Mann-Kendall change-point tests, estimating an effort-controlled decline of ~12.4% per year (IRR = 0.876) and a 50% core-use area contracting from 45 to 26 km² with a 6.5 km south-westward shift.

INDUSTRY EXPERIENCE

Financial Analysis Intern

Jinchuan Group Corporation Limited

Jun. 2024 - Sep. 2024

Jinchang, CN

Capital Budgeting Intern

State Grid Corporation of China

Jun. 2022 - Sep. 2022

Jinchang, CN

Investment Research Intern

Gansu Electric Power Investment Group Corporation Limited

Jun. 2021 - Sep. 2021

Lanzhou, CN

HONORS & AWARDS

Dean’s List	BNBU, 2022-2023, 2023-2024, 2024-2025
Bronze Award of Servant Leadership	Student Affairs Office, BNBU, 2024
Excellent Student of Environmental Awareness	Whole Person Education, BNBU, 2023

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Stata, MATLAB, SAS, SPSS, Excel VBA
- **Research software engineering:** pytest, deterministic seeding, provenance tracking, reproducibility verification, PDF parsing and OCR-assisted extraction, Git; RDF/OWL, SHACL, PROV-O, knowledge graphs, RAG, LLM-as-judge evaluation
- **Machine Learning and statistics:** PyTorch, scikit-learn, XGBoost, SHAP; panel fixed effects, 2SLS, system GMM, bootstrap inference
- **Research Data:** CSMAR, Wind, Bloomberg, WRDS, TuShare, SSE/SZSE/HKEX filings, ESG and sustainability reports
- **Languages:** Mandarin Chinese (native), English (proficient)